AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
885 A	46.02 MVA	90.0 degC	79.7 degC	30.190 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		630	mm2
Conductor type		stranded	
Conductor diameter Dc		29.2	mm
Conductor screen OD Dc_sc		31.5	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		50.9	mm
Ins outer screen OD Di_sc		52.4	mm
Insulation thickness t_ins		9.70	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
Screen type		Cu_wire	
Screen mean diameter ds		55.0	mm
Screen area As		35.0	mm2
Armour type		none	
Oversheath material		PVC	
Cable outer diameter De		60.0	mm
Rated voltage U		30.0	kV
U0 line-to-earth		18.000	kV
Frequency		50	Hz
Bonding		cross	
Formation		trefoil	
Depth to cable centre L		700	mm
Axial spacing s		60.0	mm
Ambient ground temp theta_a		30	degC
Soil thermal resistivity rho_soil		1.00	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		2.8300e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 30 degC		2.9412e-05	ohm/m
R0 at theta_max = 90 degC		3.6085e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
xs4 = [ks(8pif/R0)x10^-7]^2		12.127139	
ys skin effect factor		0.000000	
xp4 = [kp(8pif/R0)x10^-7]^2		7.761369	
yp" intermediate proximity		0.039157	
yp proximity factor		0.036083	
R at theta_max AC = R0(1+ys+yp)		3.7387e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
U0 line-to-earth		18.0000	kV
Capacitance C = er/(18 ln(Di_sc/Dc_sc)) 10^-9		0.272910	nF/m
omega = 2 pi f		314.1593	rad/s
Wd = omega C U0^2 tan(delta)		1.1112e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.00	mm2
Lay factor LFs		1.159532	
Screen Rs0 at 20 degC (with LFs)		5.7119e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_sc semiconducting screen		2.50	K.m/W
rho_tape bedding tape		6.00	K.m/W
T1_cs conductor screen		0.030167	K.m/W
T1_i insulation		0.267311	K.m/W
T1_si insulation screen		0.011556	K.m/W
T1_uwbt bedding tape under wire screen		0.019839	K.m/W
T1_raw = sum of layers		0.328873	K.m/W
T1 correction factor x1.07 (wire coverage check)		0.351894	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
T3_raw oversheath		0.069241	K.m/W
T3 x1.6 touching trefoil factor		0.110786	K.m/W
u = 2L/De		23.3333	
T4 soil		1.534109	K.m/W
Sum T = T1+T2+T3+T4		1.996789	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		79.6601	degC
Rs at theta_s		7.0511e-04	ohm/m
X sheath reactance		4.7831e-05	ohm/m
lam1' circulating current		0.000000	
lam1'' eddy current		0.025615	
lam1 = lam1' + lam1''		0.025615	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		60.00	K
Etop numerator		59.797677	
Ebot denominator		7.6230e-05	
I = sqrt(Etop/Ebot) exact		885.6845	A
I rounded TB880 GP2		885	A
MVA = sqrt(3) x U x I		46.022	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		79.6601	degC
theta_j cable outer surface		76.3154	degC
theta_a ambient ground		30.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I^2 x R		29.328072	W/m
Sheath Ws = lam1 x Wc		0.751229	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		1.1112e-01	W/m
Total		30.190416	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.156032	uH/m
Reactance X = omega L		49.0188	uOhm/m
Pos-seq resistance R1		38.3451	uOhm/m
Pos-seq impedance Z1		62.2349	uOhm/m
Capacitance C		0.272910	nF/m
Charging current Ic		1.5433	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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